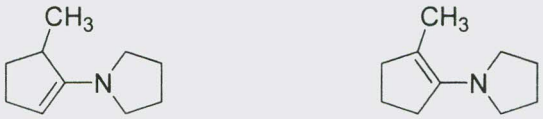
Organic Cume 10110

1. Discuss the hybridization in the C-C and C-H bonds of cyclopropane, given that the H-C-H bond angle is 118°.

2. Explain why the cyclohexane derivative shown prefers the conformation with the methyl group axial (B) rather than equatorial (A).

A **B**

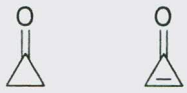
3. Predict which isomer is preferred and briefly explain why.



A **B**

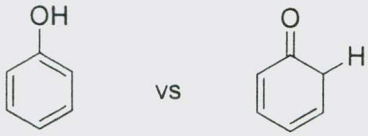
4. Draw a Newman projection of butane citing down the C2-C3 bond axis with the methyl group arranged in a +sc conformation.

5. Which is more basic? Why?



6. In the following pairs of compounds, predict which is the stronger acid and explain why.

A



B

(note: Questions 1, 2, 3, 6 are from Anslyn and Dougherty, Modern Physical Organic Chemistry, Chapters 1, 2 and 5.)